ABSTRACT
In this study, the global mortality rate attributed to air pollution between the years 1990 to 2020 will be analyzed through the application of SQL and Python. The objective of this study will be to evaluate mortality rate trends and disparities in regions while determining how different factors play into the issue. It has been observed that there has been a general decline in global mortality rates because of total air pollution and household air pollution. On the other hand, the ambient ozone pollution rate has been seen to be volatile compared to other kinds of pollution.
1. Global Trend Analysis and Forecasting.

PROBLEM STATEMENT
Although air pollution is one of the major environmental factors responsible for early deaths, its effect varies across regions and types of pollution. The aim of this study is to establish global trends, examine the effect of each pollutant on mortality rates in various regions, and determine the extent of their contribution to mortality.

Data & Methodology
WHO air pollution mortality dataset (1990–2020), containing country-level and regional-level death estimates across multiple pollution categories.

Tools Used

- SQL (data extraction, aggregation)
- Python (Pandas, Matplotlib, Seaborn)
- Visualization-based exploratory analysis

3.3 Methods

- Grouping by: GROUP BY (Country, Year, Region)
- Time series trend analysis (1990-2020)
- Comparative analysis by type of pollution:

4. Result
Global number of deaths from air pollution demonstrates a significant decline from around 34,000 deaths in 1990 to 18,000 deaths in 2020 with further declines projected to about 6,000-16,000 in

4.4 Pollution Source Relationships

A strong positive correlation exists between reductions in household air pollution and total mortality, suggesting household emissions are the primary driver of global declines. In contrast, ambient ozone pollution demonstrates weak correlation with overall trends.

4.5 Regional Disparities

Inequality in geographic distribution is present in deaths due to air pollution. The high-burden areas including Sub-Saharan Africa and South Asia have higher death rates, while developed areas have relatively low rates.

4.6 Country-Level Extremes

HIGH BURDEN:

- Solomon Islands (~28,911 deaths total)
- Afghanistan
- Bangladesh

LOW BURDEN:

- Sweden (~547 deaths)
- Switzerland (minimal household APD)
- United States (~1,285 deaths)

Insight:

> Mortality burden is heavily concentrated in developing regions, while high-income countries maintain significantly lower levels.

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5. United States Analysis

USA is regarded as an exception concerning air pollution mortality rate. The USA had around 1,285 deaths within the time frame under study, but it is still lower than those nations that registered between 10,000-28,000 deaths.

Findings

- low pollution levels within homes
- well-developed regulatory frameworks (EPA standards)
- good health care system

Despite lower overall mortality rates, residual exposure is largely due to ambient pollution caused by urban air pollution and sporadic wildfires.

6. Discussion

6.1 Strong Findings

- Global mortality from air pollution is on a downward trend
- The major factor behind the improvement is household pollution
- Ozone functions autonomously and erratically
- There are great disparities in all regions
- USA has an inherent lower burden structure

6.2 Favorable Developments

- reduction of global APD
- enhancement of household energy sources
- efficient regulation in developed nations
- achievable and verifiable success over 30 years
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6.3 Weaknesses

- linear projections cannot be guaranteed
- no standardization for population
- ozone variations remain unexplained
- comparisons between countries do not consider population

7. Conclusion

It can be seen from the analysis that there has been a consistent fall in the number of deaths attributable to air pollution across the world, mainly due to decreases in domestic emissions. Despite this improvement, there are still some regional differences with developing areas being worst hit. While the US is an exception, local pollution still poses some danger.

5. Summary Table of Key Metrics

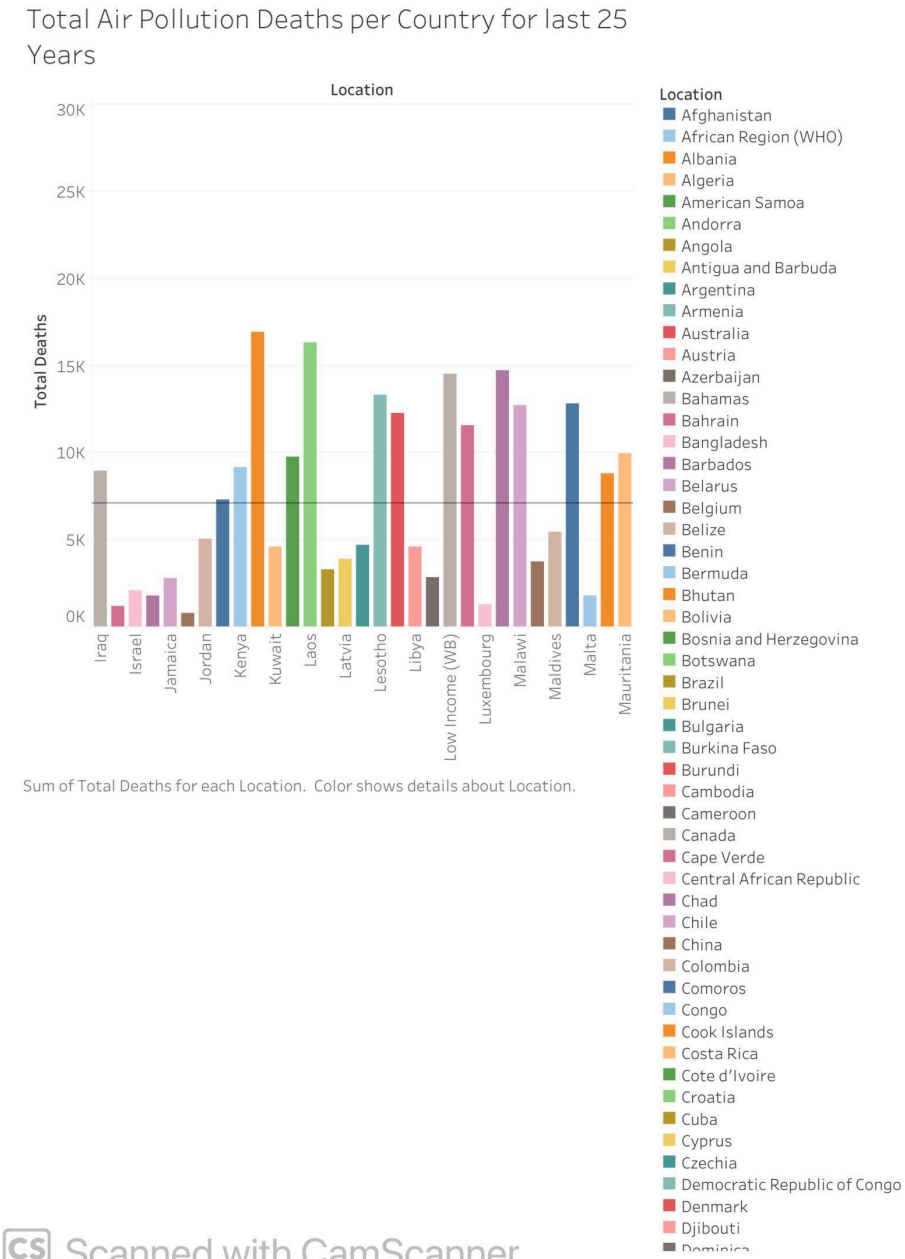
Metric	1990 Value	2020 Value	2025 Projection
Total APD	~34,000	~18,000	~16,000
Household APD	~24,000	~8,000	~6,000
Ambient Ozone	~575	~475	~460

US vs the rest of the World
US = ~1,285 total deaths over 25 years
High-burden countries = 10K–28K+ range
Global totals = declining but still much higher in developing regions

United States position:

- **Low overall burden compared to global leaders**
- Likely in the **lower decile globally**
- Not a major contributor to global APD totals

Graph#1



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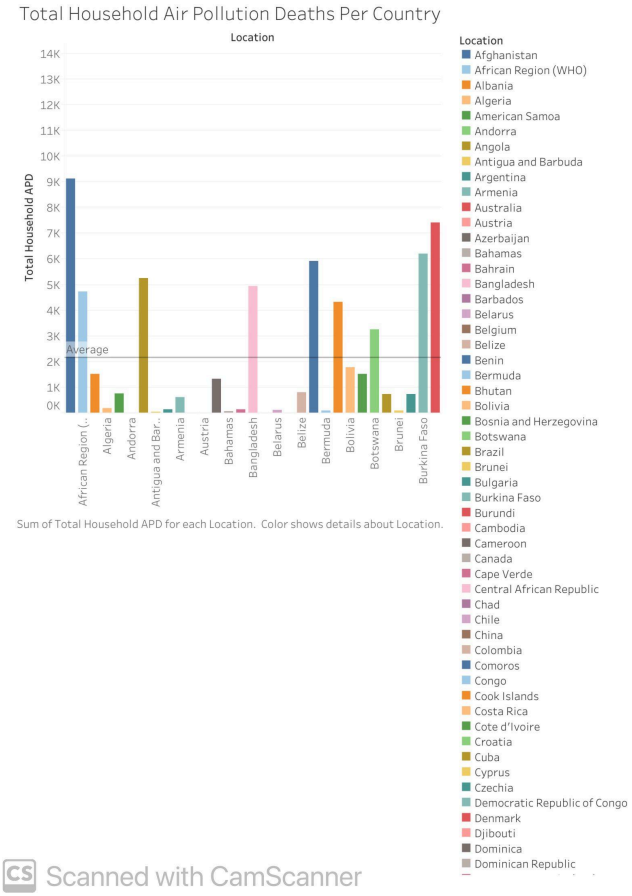
Graph#2



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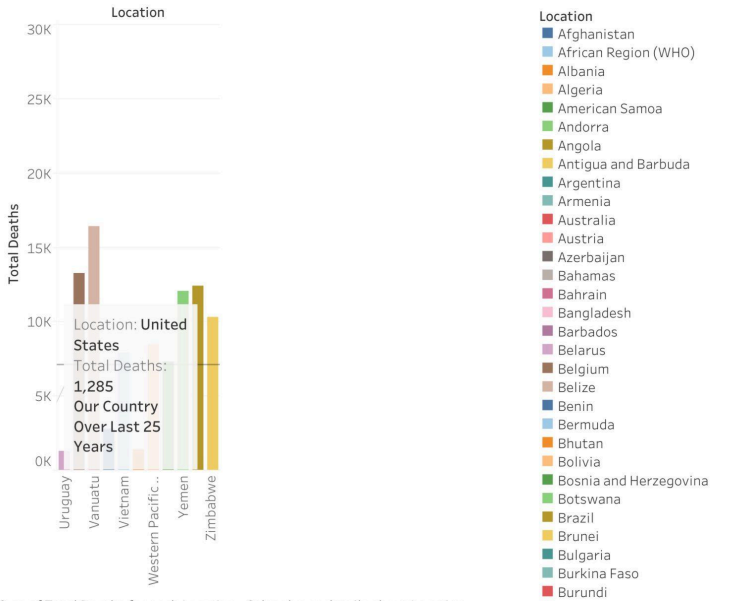
Graph #3:

Total Air Pollution Deaths per Country for last 25 Years



Sum of Total Deaths for each Location. Color shows details about Location.

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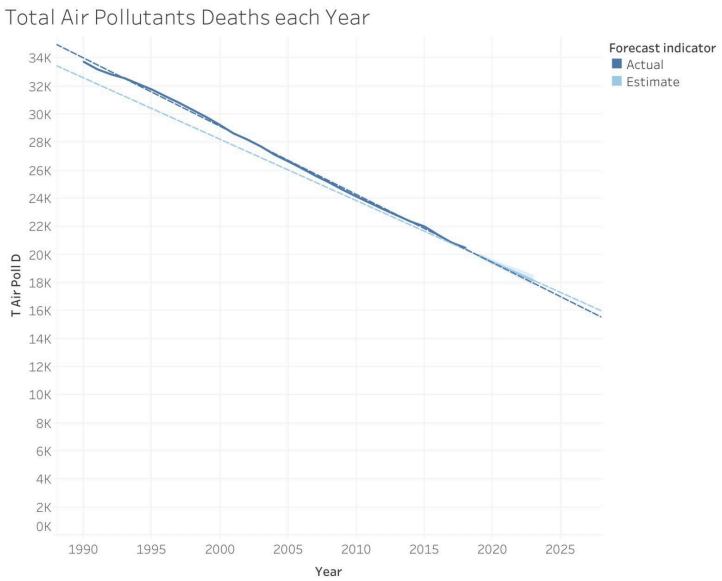
Graph #4:



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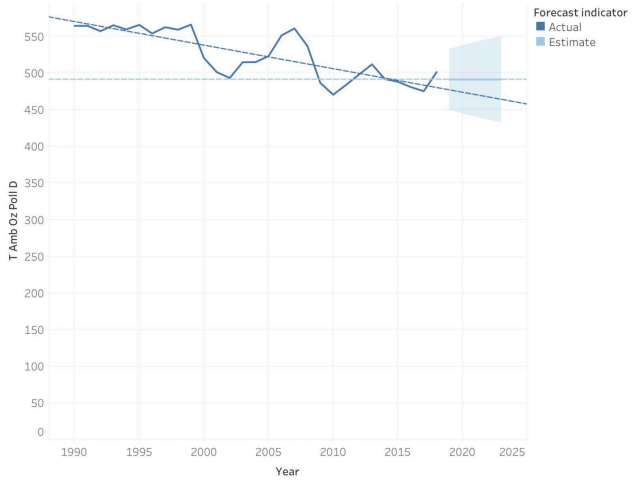
Graph #5:

Total Ambient Ozone pollutants Per Year



The trend of sum of T Amb Oz Poll D (actual & forecast) for Year. Color shows details about Forecast indicator.

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